

PPOM2 - OMM Lab 32

Introduction to Osteopathy in the Cranial Field

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** Be sure to click on sound icon for explanations on many of the following slides!

Do.
Make.
Heal.

Innovate.

Reinvent the Future.

Session Objectives

- Identify and locate key cranial landmarks.
- Describe the normal physiologic motion at the Sphenobasilar Junction (SBJ), including:
- Axes of motion
- Movement of the sphenoid and occiput during cranial flexion and extension
- Participate in a palpation session focused on whole-body sensing, starting from the feet.
- Demonstrate appropriate cranial hand contacts with proper hand/finger placement and physician ergonomics for each cranial contact (Vault Contact, Fronto-Occipital Contact, Becker and Modified Becker (Base) Contacts) on multiple partners to develop diagnostic sensitivity.
- Perform and understand the OA Decompression technique for treating the occipitoatlantal (OA) joint.

Identify the following:

Bones

A. Frontal

1. Metopic suture
2. Glabella
3. Frontal eminences
4. Supraorbital notch/foramen
5. Supraorbital margin
6. Nasion

B. Zygoma

C. Sphenoid

1. Greater wing
2. Lesser wing
3. Basi-sphenoid
4. Sella turcica
5. Clinoid processes

D. Temporal

1. Zygomatic process
2. Mastoid process
3. Squamous portion
4. External auditory meatus

E. Parietal

1. Superior and inferior temporal lines
2. Squamous portion
3. Parietal prominences

F. Occiput

1. Nuchal lines
2. Inion
3. Squamous portion
4. Basi-occiput
5. Condyles
6. Confluence of sinuses
7. Transverse sinuses

G. Ethmoid

1. Crista galli

H. Maxillae/ Mandible

Foramina:

- A. Foramen Magnum
- B. Jugular
- C. Stylomastoid
- D. Supra & Infraorbital

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Sutures

A. Lambdoid

B. Sagittal

C. Coronal

D. Parietomastoid

E. Occipitomastoid

F. Parietotemporal

G. Fronto-sphenoidal

H. Temporo-sphenoidal

Landmarks

A. Bregma

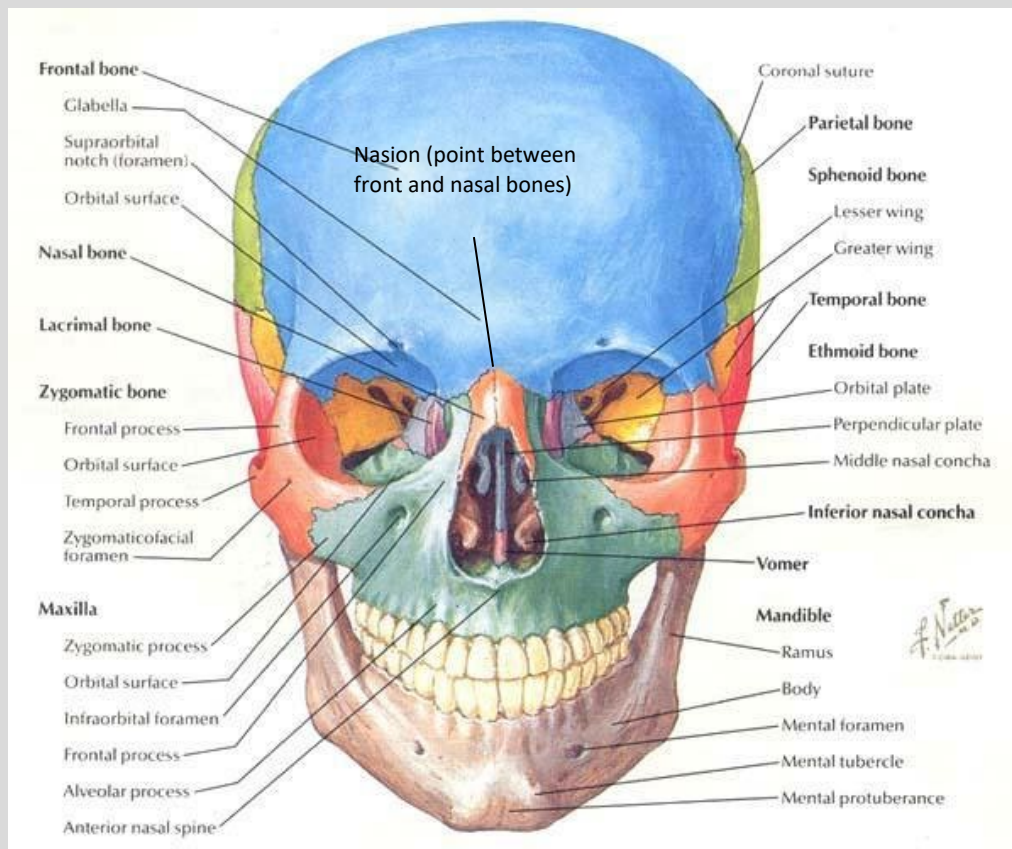
B. Lambda

C. Pterion

D. Asterion

E. Spheno-squamous pivot

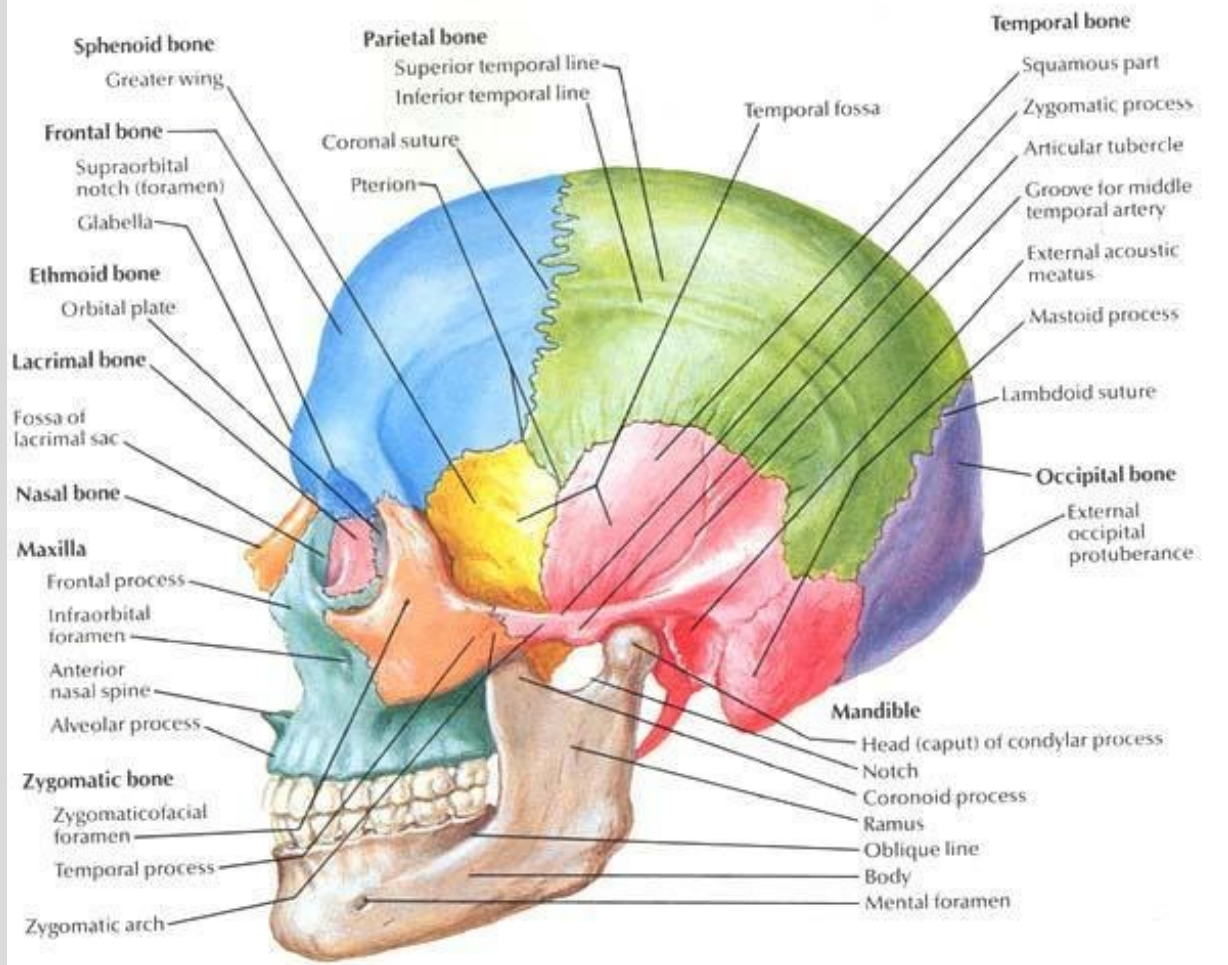
Cranial Landmarks



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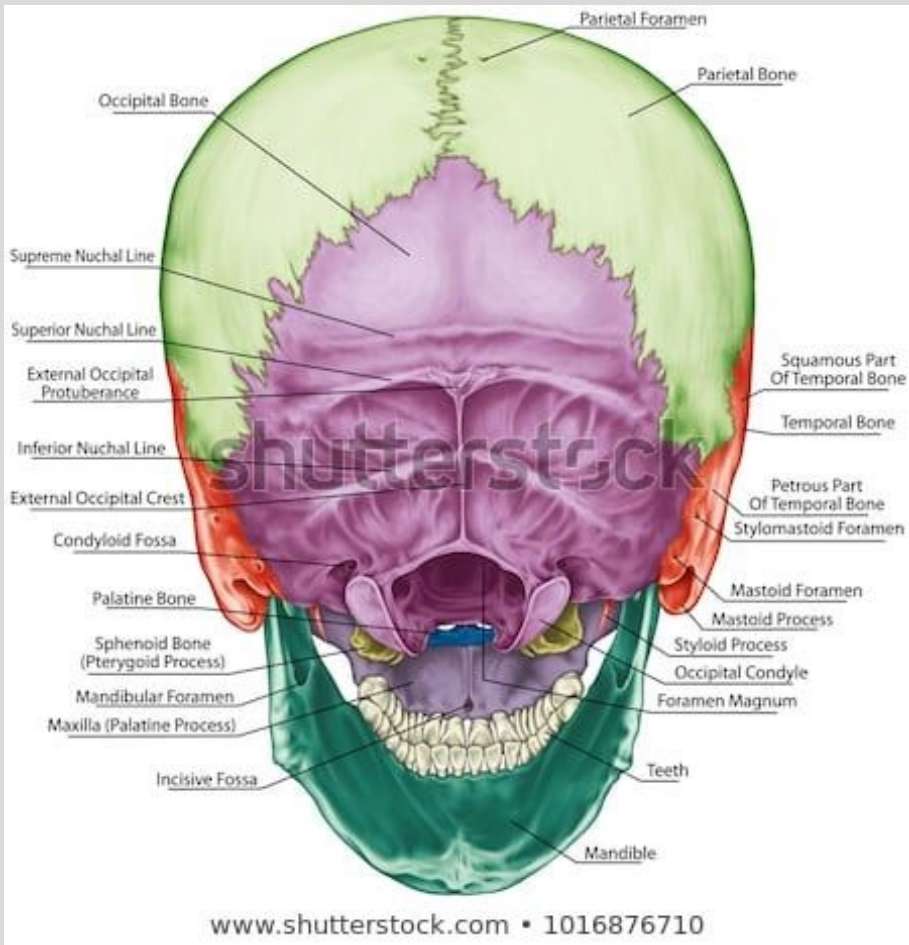


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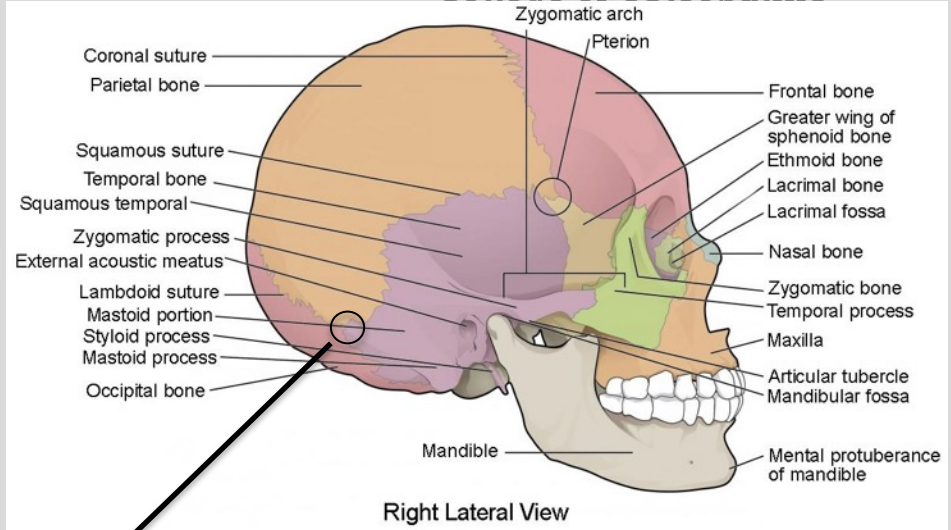
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Zygomatic arch



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The Five Phenomena of the PRM

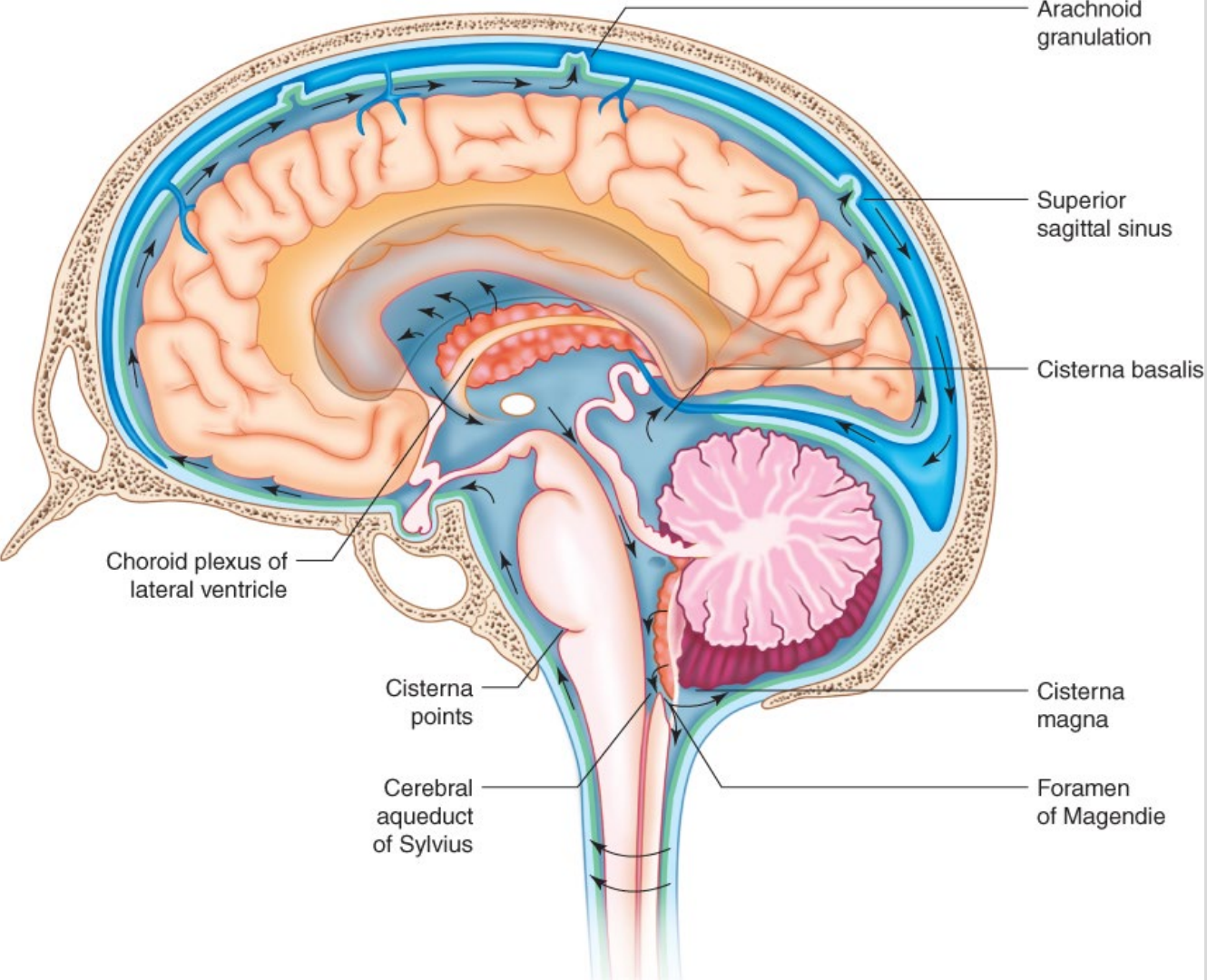
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- The inherent Motuility of the brain and spinal cord
- The Fluctuation of the cerebrospinal fluid and potency of the Tide
- The Mobility of the dural membranes
- The Mobility of the cranial bones
- The Involuntary Mobility of the sacrum between the ilia

William Garner Sutherland, Student of Andrew Taylor Still

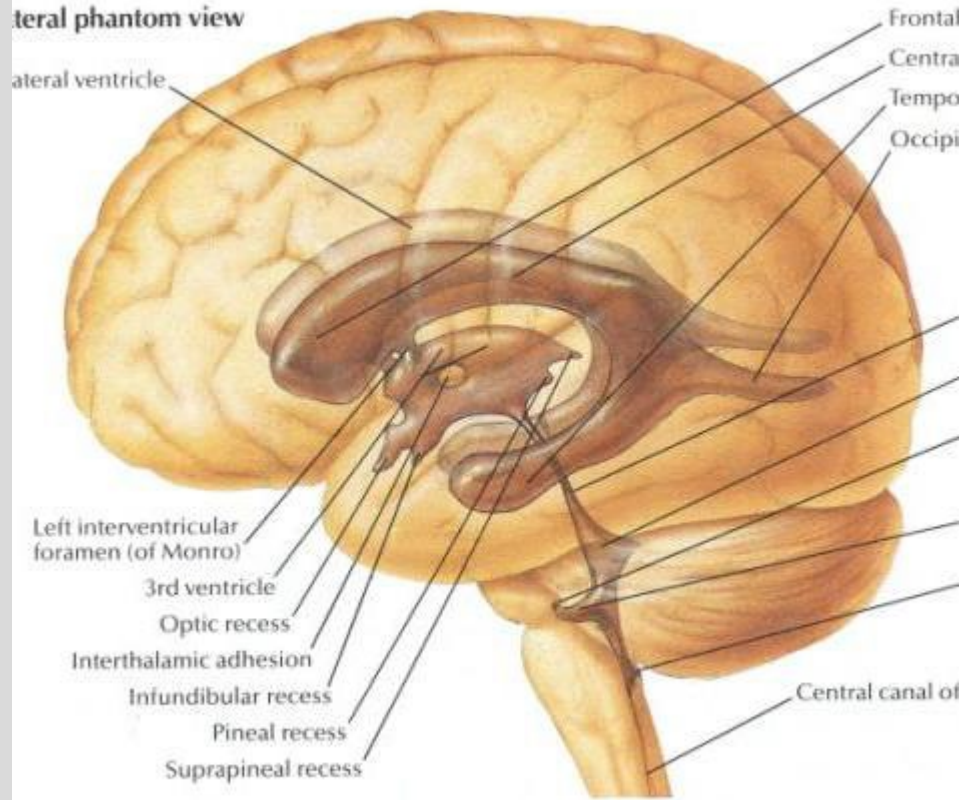




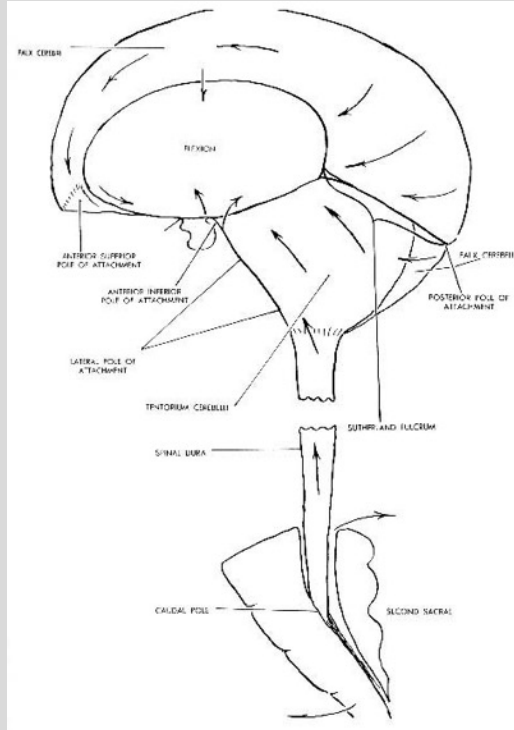
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The motility of the CNS, a coiling and uncoiling which begins as it's developmental movement. Moves synchronously with the phases of the PRM



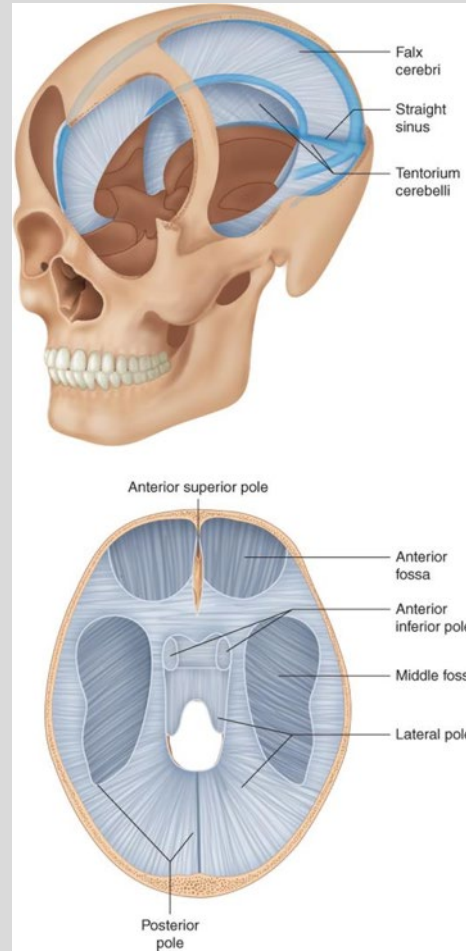
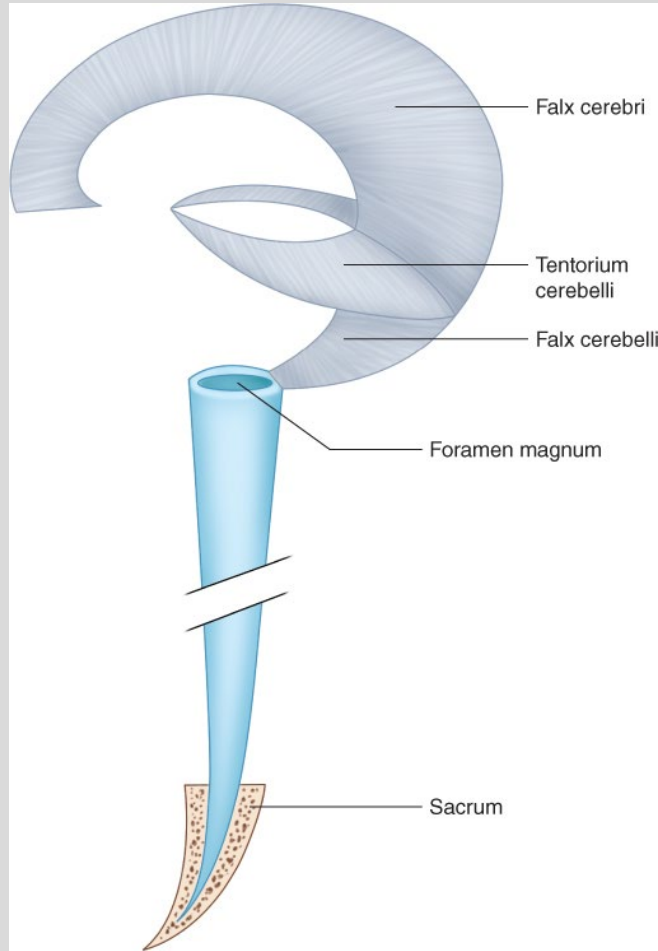
The mobility of the Dural membranes.

Act as check ligaments, guiding and limiting the movement of the cranial bones.

Motion is arcing, like that of a sickle.

Operates from a fulcrum, found along the straight sinus

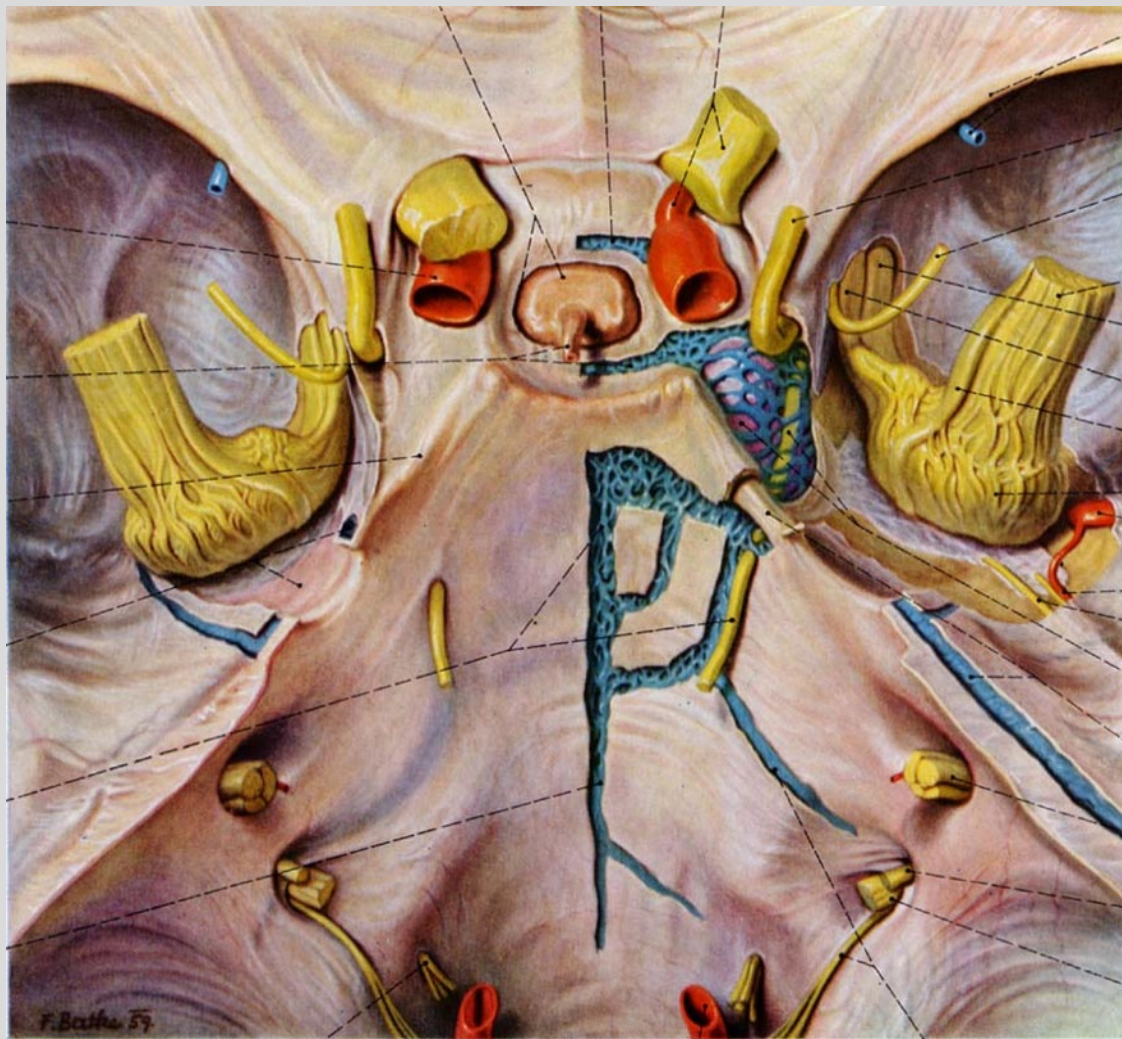
Reciprocal Tension Membrane & Core Link



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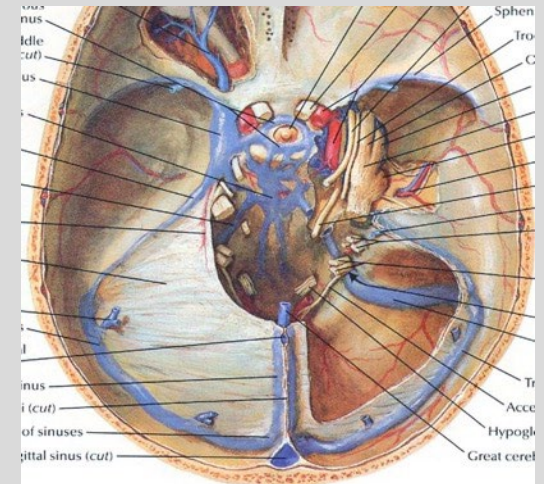
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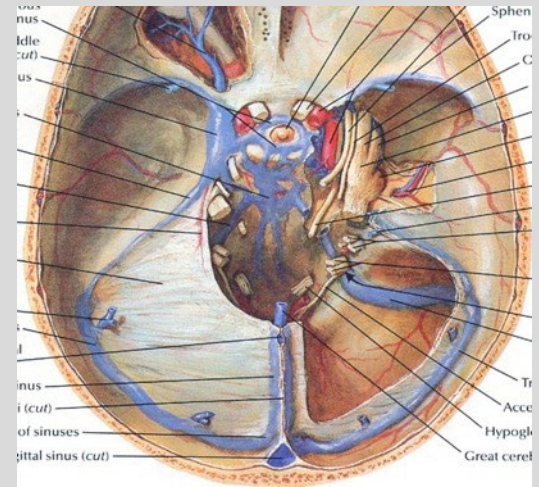


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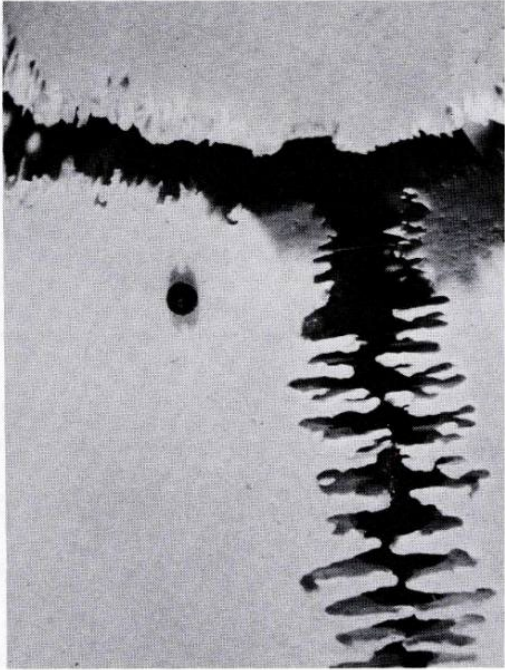


Illustration 9-2-A
Varying Lengths of Digits at Sagittal Suture

The Serrate suture:

The sagittal is the best example of a serrate suture. It's interlocking spicules allow a rocking movement, necessary to allow a widening of the parietal bones.

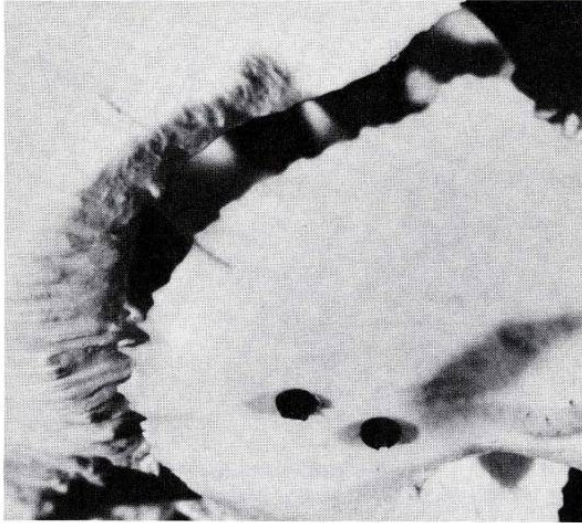
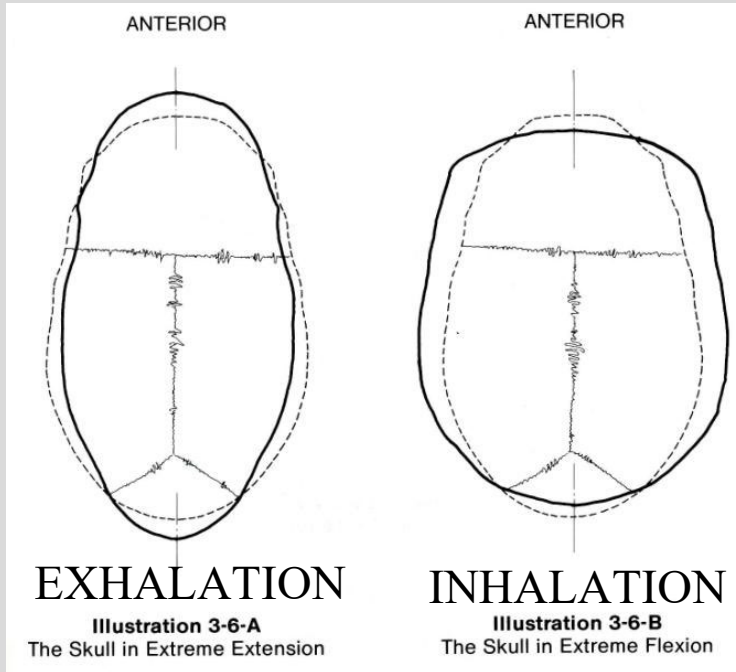


Illustration 9-2-B

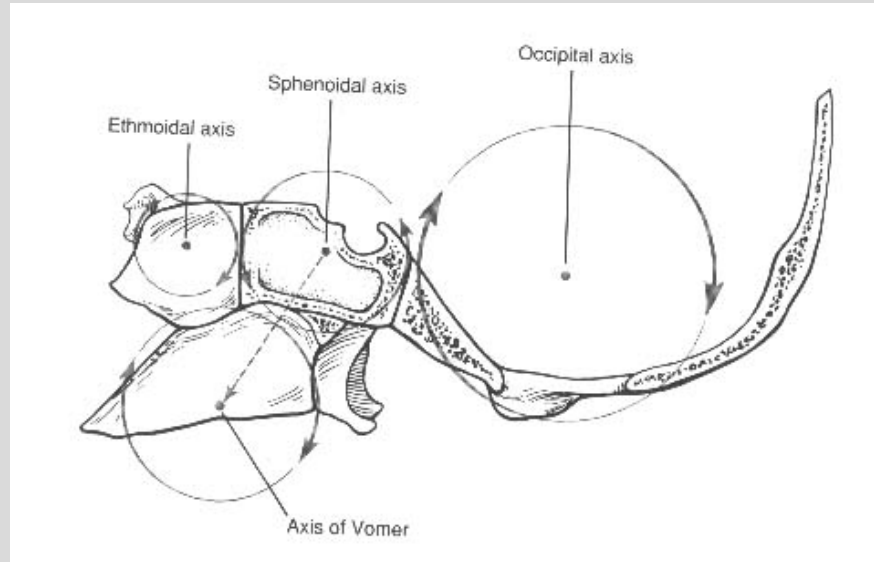
Beveled and Grooved Temporoparietal Suture

The Squamous suture:

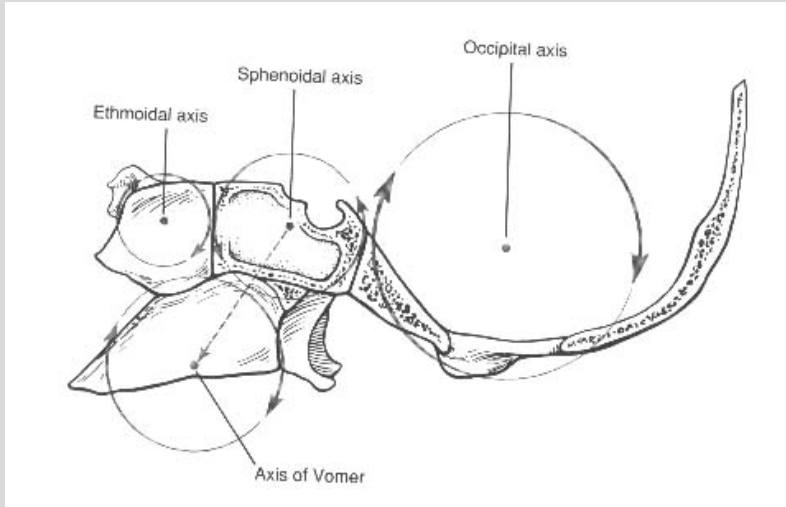
The squamous suture has marked overlapping with little interdigitating. This allows a gliding movement, necessary to accommodate the increasing and decreasing transverse diameter during inhalation and exhalation.



The movement of the primary respiratory mechanism is an alternating change in shape of the cranium. During inhalation phase, the cranium widens transversely, and narrows both A/P and vertical dimensions. The movement of the PRM may be palpated throughout the body



All midline bones rotate around transverse axes, adjacent bones moving in opposite directions, in a gearlike fashion.



Mobility of the cranial bones and sacrum:

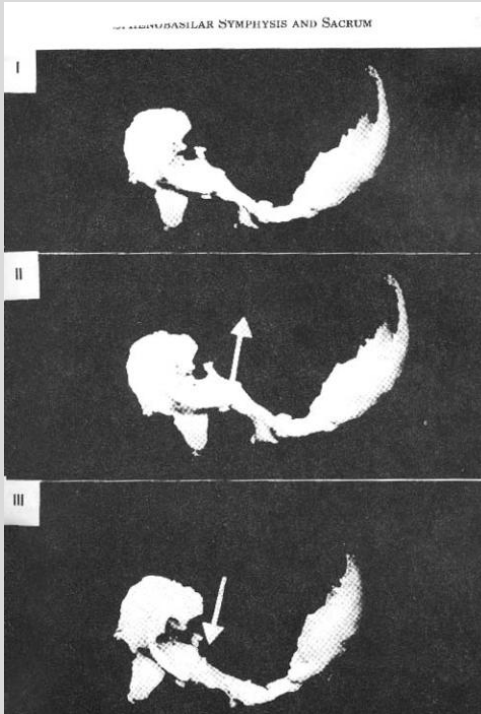
- Midline Bones
 - Flexion and Extension
- Paired bones
 - External and Internal rotation



Flexion Phase



Extension Phase



The rotation of the midline bones during inhalation and exhalation phases of the PRM is called flexion (inhalation phase), and extension (exhalation phase).

During flexion, the sphenobasilar junction moves superiorly (fig. II), and during extension, it moves inferiorly (fig. III).

This produces the vertical shape change associated with the phases of the primary respiratory mechanism.



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Inhalation phase:

Flexion midline bones,

External rotation paired bones

Sense of broadened transverse
diameter, but narrowed A/P and
vertical diameters



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Exhalation phase: Extension
midline bones,

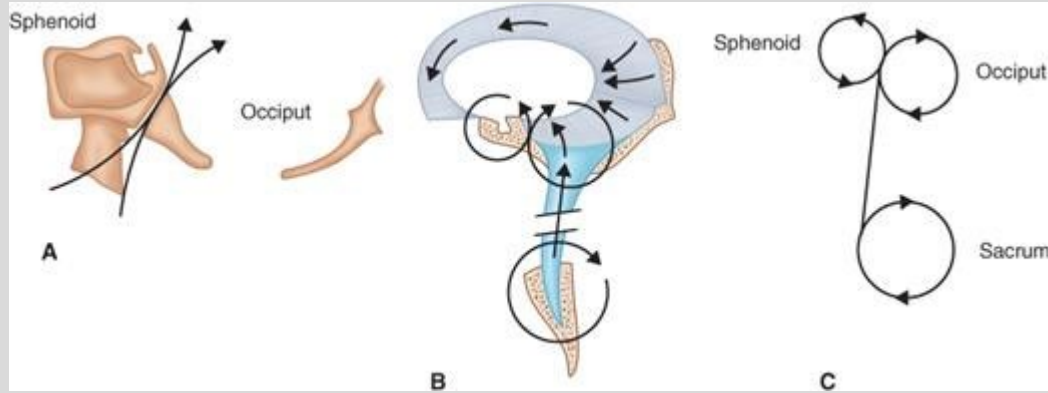
Internal rotation paired
bones

Sense of narrowed
transverse but increased A-P
and vertical dimensions

Primary Respiratory Mechanism

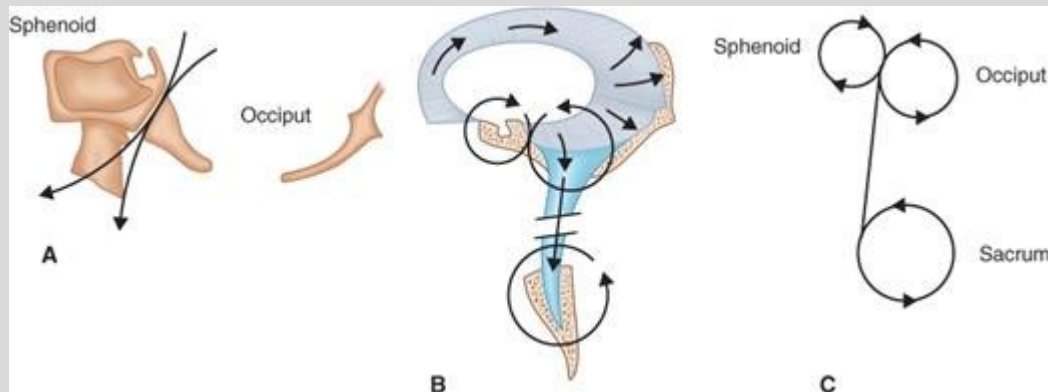
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Inhalation Phase:

Flexion of Midline bones
External Rotation of the Paired Bones
Sacral Base moves postero-superior
CNS shortens and widens (coiling)
Tentorium flattens transversely
Falx sickles and shortens A-P



Exhalation Phase:

Extension of Midline bones
Internal Rotation of the Paired Bones
Sacral Base moves antero-inferior
CNS lengthens and narrows (uncoils)
Tentorium tents transversely
Falx lengthens A-P

Do.
Make.
Heal.
Innovate.
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Beginning Sensing

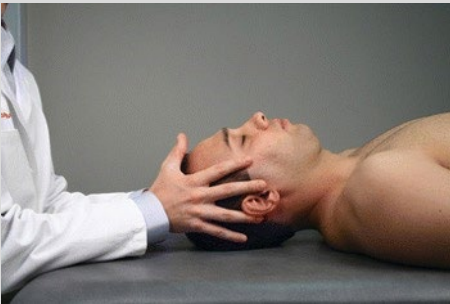
- When dealing with any patient it serves the physician well to remember to be mindful, respectful (of the patient's mechanism) & humble.
- Lifelong tips: Strive to be comfortable and respond immediately to your own discomfort- adjust yourself as often as needed.
- Be mindful of your patient's comfort. Adjust as needed, as often as needed.

Vault Contact

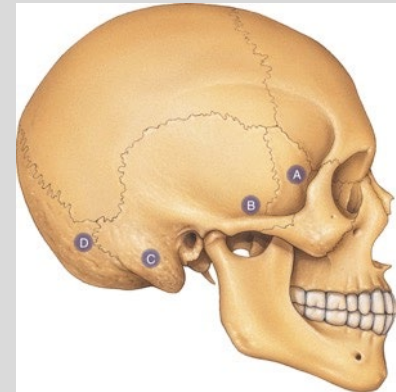
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- The patient lies supine, and the physician is seated at the head of the table.
- The physician establishes a **comfortable** fulcrum by resting both forearms on the table.
- The physician's hands cradle the patient's head, making full palmar contact on both sides.
- The physician's **index fingers** rest on the **greater wings** of the patient's sphenoid .
- The physician's middle fingers rest on temporal bone anterior to the ear.
- The physician's ring fingers rest on the temporal bone posterior to the ear.
- The physician's **little fingers** rest on the squamous portion of the patient's **occiput**.
- The physician's thumbs touch or cross each other comfortably without touching the patient's cranium.
- The physician palpates the CRI (subtle swelling and receding under your relaxed hands):
 - *Flexion/external rotation (swelling)*: coronal diameter widens, anteroposterior diameter decreases, and height decreases.
 - *Extension/internal rotation (receding)*: coronal diameter narrows, anteroposterior diameter increases, and height increases.
- The physician observes notes the amplitude, rate, and regularity of the CRI.
- The physician notes which bones, if any, have an altered amplitude, rate, and regularity.



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Fronto-occipital (A-P) Contact

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- The patient lies supine, and the physician is seated at the side of the head of the table.
- The physician places the caudad hand under the patient's occipital squama with the forearm resting on the table establishing a fulcrum.
- The physician's cephalad hand bridges across the patient's frontal bone, with the elbow resting on the table establishing a fulcrum.
- The thumb and little finger of the physician's cephalad hand rest on the greater wings of the patient's sphenoid (if the hand spread is too short, approximate the greater wings).
- The physician makes full palmar contact with both hands.
- The physician palpates the CRI:
 - *Extension/internal rotation*: coronal diameter narrows, anteroposterior diameter increases, and height increases.
 - *Flexion/external rotation*: coronal diameter widens, anteroposterior diameter decreases, and height decreases.
- The physician observes and notes the amplitude, rate, and regularity of the CRI.
- The physician notes which bones, if any, have an altered amplitude, rate, and regularity.
- The physician directs palpatory sense at the SBS, determining whether there is any preferred motion of the sphenoid and the occiput.



<http://arktos.nyit.edu/login?url=https://meded.lwwhealthlibrary.com/content.aspx?sectionid=123844148&bookid=1629#123848666>



Becker Hold

- Gently contact with your thumbs at the sphenoid greater wings and the rest of the fingers gently cradling the temporal bones and occiput.
- Observe the cranial rhythmic impulse and evaluate for rate, amplitude, and symmetry.



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Temporal contact (Becker variation)

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- Hands gently cradle the posterior cranial fossa.
- Let thumb tips slide antero-inferior to the mastoid tips so as not to compress them.
- Palpitorally observe as in previous slides.



OA Decompression

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*Occipitoatlanto (C0-C1) is the first cranial joint formed in the body

→ Absorbs the initial forces of labor and delivery

→ **NOTE:** *This particular treatment technique is not to be used on infants and small children (see the following slide with the C1 ossification centers schematic). For pediatric patients who have reached the age of approx. 8 to 10 years, one may consider utilizing this treatment technique approach.*

Landmarks:

- Occipital condyles & atlas
- Inion & C1 posterior tubercle

Hand placement:

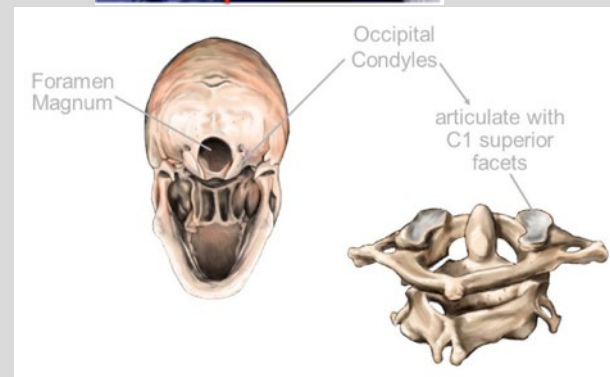
- Can start with middle finger on inion, then one will slide the finger down until at area of the C1 posterior tubercle (If at spinous process of C2, you have gone too far)
- Pt. flexes head (not the neck – pt. tries to tuck chin to chest), and doctor will curl finger toward glabella, stabilizing the atlas via the projected contact towards the C1 posterior tubercle
- Other hand: gently apply a slight pressure to stabilize patient's head in flexion

Treatment:

- Hold until muscle and soft tissues begin to relax, and one has a sense of the occipital condyles becoming decompressed from the receiving condyles of the C1 superior facets. Reassess.

Clinical Implications:

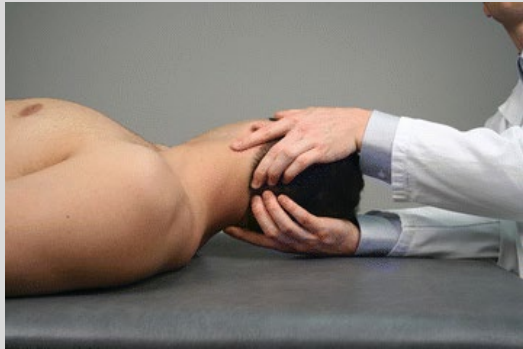
- Swallow (CN IX-X)
- Visceral disturbance (CN X)
- SCM and trapezius (CN XI)
- Cranio-cervical disturbance



Technique source: Teachings in the Science of Osteopathy, William Sutherland Pg 230

OA decompression, alternative

- The patient lies supine, and the physician is seated at the head of the table with both forearms resting on the table, establishing a fulcrum.
- The physician places the pads of both middle fingers on the posterior aspect of the cranium and slides these fingers down the occiput until the fingers are against the posterior arches of the atlas (**Figs. 18.17 to 18.19**).
- The physician applies caudad pressure with both middle fingers to separate the facets from the condylar parts.
- While the physician maintains this caudad pressure, the patient tucks the chin into the chest, making sure *not* to flex the neck (this is the nodding movement that occurs at the occipitoatlantal joint).



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OA decompression, alternative

- This motion carries the occipital condyles posteriorly, tenses the ligaments in the region, and stretches the contracted muscles in the occipital triangle.
- The physician maintains this position while the patient holds one or more deep inspirations to their limit. This will enhance articular release.
- The rate and amplitude of the CRI, as it manifests in the basioccipital region, are retested to assess the effectiveness of the technique. Occipitoatlantal motion testing can also be assessed for normalization.



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The following 3 exercises are **optional** to do before class. We will introduce these concepts in lab while you are learning how to sense more subtle motions of the living mechanism. They are here to serve as a guide for practice and betterment on your own.

Sensing Inherent Motion

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Exercise 1

Sensing yourself

- Patience
- Relaxed
- Courage
- Confidence
- Humility



Sensing Inherent Motion

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Exercise 2 - Sensing the whole

- Posture
- Center
- Breath



Sensing Inherent Motion

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- Exercise 3 – Sensing Inherent Motion
- Robert Fulford, DO
- Proprioception

Session Objectives

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1. Identify and locate ALL the various cranial landmarks and give examples of clinical implications of sutural restriction, cranial nerve entrapments, dural restriction and venous sinus congestion which are components of cranial somatic dysfn.
2. Compare, contrast and demonstrate different cranial contacts (holds) for dx and tx. Be able to explain proper hand/finger placement, body attitude/receptivity, comfort etc
3. Review & describe the 5 phenomena of primary respiratory mechanism (PRM) and cranial rhythmic impulse (CRI) in OCF and make your first attempts to sense them.
4. Practice palpating the PRM on others
5. Explain what happens to midline and paired parts during both phases of the PRM (inhalation/exhalation), and how that should feel during the various hand contacts (holds)
6. OA decompression treatment

Lab Feedback Form:

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REQUIRED READING & VIDEO LINKS

This is the first cranial labs. The material builds on itself. Please do yourselves a favor and put the work in early to try to understand and appreciate this very clinically useful aspect of osteopathy.

[Chp 103](#) Anatomic Considerations

[Chp 104](#) 2 pages: Cranial Motions and Dysfunction (up to, not including Torsion)

[Chp 105](#) 1 page: Cranial Dx and Tx (Read up to not including Principles of Treatment)

[OPTIONAL Chp 106](#): Clinical Correlations (OPTIONAL if you have time)

If you have time you can get ahead for next labs by reading the rest of Chp 104-6

KALTURA VIDEOS:

Cranial Landmarks <http://www.kaltura.com/tiny/iegx2>

Vault contact <http://www.kaltura.com/tiny/stue7>

Frontal-Occipital (A-P) contact <http://www.kaltura.com/tiny/pp8se>

Becker contact and Temporal contact variation <http://www.kaltura.com/tiny/oz141>

OA Decompression https://nyitmedicine.mediaspace.kaltura.com/media/OA+Decompression/1_loeacfh0